



Spatial patterns of conifer regeneration following Mountain pine beetle outbreak (*Dendroctonus ponderosae*) and mixed-severity wildfire: A point process model of disturbance interactions in the southern Selkirk Mountains, British Columbia

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ABSTRACT

Mountain pine beetle (MPB) outbreaks and wildfire increasingly overlap across conifer forests of western North America, yet few studies have examined how prior beetle mortality shapes post-fire regeneration at fine spatial scales. We investigated seedling intensity of seven conifer species across three 4-ha plots within the 2015 Mt Midgeley Wildfire burn site in the Darkwoods Conservation Area, British Columbia, where mixed-severity fire followed MPB outbreaks by 1–11 years (median 7 years). Disturbance covariates were derived from classified Landsat time-series imagery using the differenced Normalised Burn Ratio (dNBR) for burn severity and the differenced Normalised Moisture Index (dNMI) for red-stage beetle mortality, both validated against field reference plots. Grey-stage MPB mortality was derived as the union of annual Landsat NDMI red-attack classifications from 2005 to 2011, identifying pixels in which beetle-killed canopies had ≥ 4 years to lose needles and collapse before the 2015 fire. Among 3886 mapped seedlings, spatial point pattern analysis revealed significant clustering for all species, with intensity governed by fire severity, distance to seed source, topography, and wind. Inhomogeneous Cox process models showed that areas of overlapping beetle mortality and high burn severity had significantly reduced lodgepole pine regeneration, while other conifers responded primarily to fire severity and seed source proximity. Results suggest that as beetle-fire intervals shorten under climate change, divergent regeneration trajectories between host and non-host species may alter successional outcomes and challenge the resilience of pioneering lodgepole pine cohorts.

1. Introduction

Interacting disturbances, including overlapping pulses of insect outbreak, wildfire, drought, and wind, are reshaping forested biomes worldwide. Where these events compound on intervals shorter than the recovery time of dominant species, post-disturbance trajectories diverge from the predictions of single-disturbance models (Buma, 2015; Burton et al., 2020; Seidl et al., 2017). Across temperate North America, bark beetle outbreaks and wildfire are increasingly the dominant pair, with their joint footprint expanding under warming climate (Raffa et al.,

2008). The forests of British Columbia's southern interior have entered such a period of intensified compound disturbance.

In recent decades, the forests of British Columbia's southern interior entered a period of intensified disturbance. Mountain pine beetle (MPB) outbreaks impacted millions of hectares of lodgepole pine (*Pinus contorta* var. *latifolia*) across the province's interior plateau between the late 1990s and mid-2010s (Raffa et al., 2008; Taylor et al., 2006; Westfall and Duthie-Holt, 2021; Westfall and Ebata, 2011), while wildfire seasons have brought longer, more frequent, and more severe events (Daniels et al., 2020; Filmon, 2004; Hamann and Wang, 2006; Littell

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et al., 2009; Loehman et al., 2011; Waring and Coops, 2016). Where these disturbances overlap in space and time, their interaction may alter post-fire regeneration in ways that neither disturbance alone would produce (Buma, 2015). Whether such interactions constitute “linked” or compounded effects or simply sequential disturbances with independent ecological consequences remains an open question (Burton et al., 2020), one that depends on the temporal stage of beetle attack, species-specific regeneration traits, and local site conditions (Harvey et al., 2014; Stevens-Rumann et al., 2015). These concerns are most acute in the fire-prone, pine-dominated forests of southern interior British Columbia and western Alberta, where the geographic extent of both MPB outbreaks and wildfire has been expanding under warming conditions (Jia et al., 2016; Safranyik et al., 2006; Wulder et al., 2006a, 2006b).

Numerous studies have explored the effects of pre-fire MPB outbreaks on fuel profiles and the ecological severity of subsequent wildfire (Harvey et al., 2014; Hicke et al., 2012; Jenkins et al., 2012; Meddens et al., 2012; Page et al., 2013; Perrakis et al., 2014; Simard et al., 2011). Findings tend to be mixed. Reports indicate heightened crown fire activity, faster initial rates of spread (Perrakis et al., 2014), and deeper charring (Talucci and Krawchuk, 2019) in MPB-killed stands compared with unaffected lodgepole pine stands. More recent work has examined whether beetle-fire sequences alter the distribution, abundance, and establishment rates of seed banks across gradients of MPB and wildfire history (Harvey et al., 2016, 2014; Talucci et al., 2019a, 2019b). These results, too, remain inconclusive. Variation in outcomes has been attributed to the unpredictability of fire behaviour during green-stage MPB attack (Wulder et al., 2006b), uncertainties surrounding the viability of serotinous cones following crown and surface fires, and the resilience of ground-level seed banks (Hicke et al., 2012; Schoennagel et al., 2012; Simard et al., 2011).

A key source of this variability is the temporal stage of MPB attack, which fundamentally transforms fuel structure and loading between the green, red, and grey phases (Harvey et al., 2014; Stevens-Rumann et al., 2015). In the first 12–24 months following attack, infested trees retain red needles as sapwood and foliar moisture decline (Amman, 1983, p. 198; Henigman et al., 1999; Jenkins et al., 2012; Jolly et al., 2012; Page et al., 2013). During this red stage, dead needle retention exceeding 50% adds fine fuel to the canopy (Simard et al., 2011), while increased light penetration and wind exposure accelerate the drying of surface fuels and fuel ladders (Harvey et al., 2014; Meddens et al., 2012). The grey stage follows as remaining needles and heavy snags fall to the forest floor, contributing downed logs, accumulating litter, and opening conditions for sapling and grass establishment (Carroll et al., 2003; Donato et al., 2013; Meddens et al., 2013; Schoennagel et al., 2012). The ecological consequences of fire burning through beetle-affected stands thus depend heavily on when, relative to the outbreak trajectory, ignition occurs.

Beyond fuel structure, temporal dynamics also govern the regeneration processes and seed bank viability of dominant conifers in beetle-fire sequences (Bell et al., 2014; McCullough et al., 1998). For lodgepole pine, serotiny mechanisms are closely tied to fire-dependent triggers, with regeneration favoring pyriscent (fire-cued) cues over xeriscent (drought-cued) or senescence pathways (Buma, 2015; Buma and Wessman, 2011; Despain et al., 1996; Lamont et al., 1991; Talucci et al., 2019a; Teste et al., 2011). Fire activates the release of large seed pulses from viable canopy stores. In grey-stage stands, however, where canopy structure has collapsed, higher rates of xeriscent dispersal occur from fallen ground stores that are vulnerable to overheating during surface fires, prolonged flame residence, and increased predation.

For non-host conifers, the risks posed by surface fire vary with species-specific traits. Western larch (*Larix occidentalis*), Engelmann spruce (*Picea engelmannii*), and western hemlock (*Tsuga heterophylla*) produce small cones with light, papery seeds whose ground-level banks are readily consumed in surface fires. Successful dispersal in these species relies on wind, making topographic position and distance to seed source critical determinants of regeneration. Thicker-barked Douglas-fir (*Pseudotsuga menziesii*), with heavier seeds, depends less on wind than on

post-fire proximity to surviving seed trees and favorable microsites. The largest-coned species, ponderosa pine (*Pinus ponderosa*) and western white pine (*Pinus monticola*), produce thickly coated, single-winged seeds whose dispersal beyond wind thresholds is primarily zootic (Loehman et al., 2011; Owens, 2005; Owens and Molder, 1977; Toffin et al., 2018; Vander Wall, 2023; West, 1968).

In subalpine and montane forests where disturbance intervals are shortening, topography exerts a strong influence on regeneration. South-facing slopes experience greater fire severity and post-fire moisture stress, effects likely to intensify under warming climates, while north-facing slopes and areas of geomorphic sheltering offer conditions where regeneration may persist despite changing disturbance regimes (Hoecker et al., 2020). Such topographic effects are especially pronounced at species range margins (Meidinger and Pojar, 1991) and have been documented in the establishment patterns of western hemlock (Grossnickle and Russell, 2010), ponderosa pine and Douglas-fir (Chambers et al., 2016; Lerch et al., 2016; Sieg et al., 2017; Ziegler et al., 2017), and western larch (Oswald and Neuschwander, 1995; Sparks et al., 2018). Topographic position mediates both the spread of MPB attack and post-attack conifer recovery: cooler, moister aspects support higher host density and slower beetle development, while warmer, drier aspects experience faster outbreak progression and slower seedling establishment (Hadley, 1994). This dual role of topography as both disturbance modifier and regeneration filter motivates our explicit inclusion of aspect, slope, and terrain ruggedness as covariates.

This study investigates three hypotheses grounded in the fire strategist classification system of Rowe (Rowe, 1983) and subsequent refinements (Agee, 1998; Dobrowski et al., 2015; Hamann and Wang, 2006; Mathys et al., 2018). This framework categorizes conifers as fire-resisters, fire-avoiders, fire-evaders, or fire-invaders according to their seed production, dispersal mechanisms, establishment requirements, and adult resilience (Table 1).

Fire-resisters (e.g., mature Douglas-fir and ponderosa pine) survive low- and moderate-severity fire through thick bark and elevated crowns. Fire-avoiders (e.g., Engelmann spruce and western hemlock) are killed by most fire but persist in fire-protected refugia and recolonise slowly via wind dispersal. Fire-evaders (e.g., lodgepole pine) rely on serotinous cones to release seed pulses immediately after fire. Fire-invaders (e.g., western larch and western white pine) colonise newly burned areas from off-site seed sources via long-distance wind or animal dispersal (Rowe, 1983).

The hypotheses explore how varying disturbance histories of beetle outbreak and wildfire shape seedling patterns among host and non-host species across topographic contexts. First, we anticipate significantly lower seedling intensity in areas of high-severity fire among fire-avoiders and fire-resisters than among fire-evaders and fire-invaders. Fire-avoiders and fire-resisters depend on surviving seed trees or off-site dispersal, while fire-evaders regenerate from serotinous canopy seed pulses and fire-invaders from long-distance wind and animal dispersal (Rowe, 1983; Stevens-Rumann et al., 2015). Second, areas of red- and grey-stage beetle outbreak are expected to exhibit lower seedling intensity among host species (*Pinus spp.*) than among non-host species. Beetle attack consumes the standing seed source on which fire-evading *Pinus* contorta depends (Harvey et al., 2014; Talucci et al., 2019a). Third, areas subjected to both high-severity fire and beetle outbreak are expected to demonstrate lower seedling intensity among host species than among non-host species, with these patterns mediated by topographic features at the microsite scale. Topography modifies this combined effect through its control on fuel moisture, fire residence time, and microsite suitability (Harvey et al., 2016; Stevens-Rumann et al., 2015).

Most post-fire regeneration studies tally seedlings within distributed small plots, capturing mesoscale variation but obscuring the fine-scale pattern in which seedlings actually establish. Stem-mapping every seedling within larger plots reverses this trade-off: it resolves spatial structure at metres rather than kilometres, allowing tests of the

Table 1

Regeneration traits, fire-response strategies, and seed dispersal characteristics of seven conifer species in the southern Selkirk Mountains study area, classified according to the fire strategist framework of Rowe (1983).

Species	Mature bark	Young bark	Sexual maturity	Fire resistance	Seed dispersal	Seed production ^a (n/gm, length x width)	Seedbed preferences	Fire strategy
Douglas fir (<i>Pseudotsuga m.</i>)	Very thick	Thick	12–15 yrs	High when mature	3% > 80 m	Bumper crops / 2–11 yrs; 103/gm, 5.2 × 3.1 mm	Mineral and organic soils, limited shade intolerance	Fire-resister
Engelmann spruce (<i>Picea engelmannii</i>)	Thin	Very thin	15–40 yrs	Very low	50% < 30 m 20% < 60 m	Bumper crops / 3–5 yrs; 439/gm, 2.8 × 1.7 mm	Moist, mineral soil, partial shade tolerance	Fire-avoider
Lodgepole pine (<i>Pinus contorta</i>)	Very thin	Very thin	5–15 yrs	Low	50% < 18 m, 20% < 40 m	Bumper crops / 1–3 yrs; 346/gm, 3.4 × 1.9 mm	Mod-high severity burn, mineral soils in full light	Fire-evader
Ponderosa pine (<i>Pinus ponderosa</i>)	Very thick	Thick	4–8 yrs	Very high	50% > 25 m, 20% > 60 m	Bumper crops / 2–8 yrs; 20/ gm, 8.9 × 5.7 mm	Mineral soils or ash, in full light	Fire-resister
Western hemlock (<i>Tsuga heterophylla</i>)	Medium	Thin	25–30 yrs	Low	50% < 90 m, 20% < 120 m	Bumper crops / 3–4 yrs; 494/gm, 3.1 × 1.9 mm	Shade tolerant, sensitive to frost and drought	Fire-avoider
Western larch (<i>Larix occidentalis</i>)	Very thick	Thick	25–45 yrs	Most resistant	50% > 45 m, 20% > 110 m	Annual, bumper / 5 yrs; 283/gm, 4.4 × 2.2 mm	High severity burn, mineral soils in full light	Fire-invader
Western white pine (<i>Pinus monticola</i>)	Thin	Very thin	20–30 yrs	Low	< 32 km (zootic)	Bumper crops / 2–4 yrs; 52/ gm, 6.8 × 3.9 mm	Moderate-high severity burn patches	Fire-invader

^a Seed morphology were compared according to longitudinal and cross-sectional dimensions measured from micropyle to chalazal end, and between upper (abixial) and lower (adaxial) surfaces of a winged female ovule fertilized by male pollen grain and documented within interior British Columbia (Allen, Owens, 1972; Kolotelo, 1997; Kozłowski, 1972; Owens and Molder, 1977; Singh, 1978).

mechanisms by which interacting disturbances are hypothesised to govern fine-scale variability in regeneration. Four mechanisms are tested in what follows: distance to conspecific seed source, microsite topography, dispersal mode, and the substrate created by beetle-killed stems.

2. Materials and methods

Scientific studies combining remote sensing of wildfire and beetle outbreak with point pattern analysis to investigate fine-scale spatial patterns of post-disturbance regeneration across multiple conifer species remain scarce (Chambers et al., 2016; Talucci et al., 2019a; Talucci and Krawchuk, 2019). This study applied stem-mapping methods and supervised classification of multispectral imagery to examine conifer seedling patterns established following the mixed-severity Mt Midgeley Wildfire of 2015 in areas affected by a pre-fire MPB outbreak from 2004 to 2014 (MFLNRO, 2015). The analysis proceeded in two stages. In the first, we mapped disturbance histories and topographic covariates using field sampling, ground truthing, digital terrain models, public geospatial datasets, and multispectral vegetation-differenced indices. In the second, we sampled seedling locations within three 4-ha plots and fitted inhomogeneous log-Gaussian Cox process models with distance-weighted spatial covariates derived from the first stage to assess species-specific patterns of seedling density and intensity across disturbance gradients.

2.1. Study site

The Mt Midgeley Wildfire of August 2015 burned 479.77 ha of forest along the southeast perimeter of the Darkwoods Conservation Area, British Columbia (MFLNRO, 2015). The Landsat dNBR classification of the burn extent (see §2.3) resolves the mixed-severity character of the fire across four classes: 48.15 ha of high-severity burn (10.0%), 114.12 ha of moderate severity (23.8%), 192.69 ha of low severity (40.2%), and 124.20 ha of unburned forest (25.9%), within a total footprint of 479.16 ha. Class proportions agree closely with the BC Wildfire Service classification for the low and unburned classes; high-severity area is approximately twice the BC estimate, and moderate-severity area correspondingly smaller, reflecting the higher fire-effect sensitivity of the dNBR index at the canopy scale. Mountain pine beetle activity within the burn perimeter was first detected in 2004 (Buston and MacLachlan, 2018), with attacks of light to severe intensity

accumulating across 231.61 ha (48.2%) by the end of 2014 (Westfall et al., 2020).

The site spans elevations from 830 to 1744 m a.s.l. across three biogeoclimatic subzones: the Yahk Wet Mild Engelmann Spruce-Subalpine Fir zone (ESSFwm4, 207.63 ha, 43.28%), the Ymir Moist Warm Interior Cedar-Hemlock zone (ICHmw4, 84.72 ha, 17.69%), and the West Kootenay Dry Warm Interior Cedar-Hemlock zone (ICHdw1, 187.42 ha, 39.07%) (MacKillop, 2018; Meidinger and Pojar, 1991). Soils are developed on fine-grained metasedimentary parent materials of argillite, sandstone, and mudstone, with sandy loam textures overlying medium to coarse colluvial and glaciofluvial deposits, occasional morainal debris, and granodiorite outcrops. Surface soils are classified as Humo-Ferric Podzols, Dystric Brunisols, and Gleysols (Barton et al., 2004; Cochrane, 2007).

Pre-fire forest composition was dominated by lodgepole pine, with co-dominant Engelmann spruce at upper elevations (Beaudoin et al., 2018) and mixed mosaics of Douglas-fir, western larch, ponderosa pine, western hemlock, and western white pine at lower elevations (MacKillop, 2018).

A digital elevation model at 1 m resolution was obtained from the High Resolution Digital Elevation Model repository of Natural Resources Canada (Natural Resources Canada, 2017). The DEM was processed with Gaussian smoothing, interpolation, and a sink-filling algorithm (Wang and Liu, 2006) to derive rasters of aspect, slope, topographic wetness index (TWI), and terrain ruggedness index (RIX) (Parker, 1982). Raw aspect (0–360°) was decomposed into northness = $\cos(\text{aspect} \times \pi/180)$ and eastness = $\sin(\text{aspect} \times \pi/180)$, with northness retained as the covariate term following Beers et al. (1966) and Pierce et al. (2005). This avoids the circular discontinuity at 0°/360° that would otherwise distort regression coefficients. Terrain ruggedness was tested at multiple resolutions; a final resolution of 30 m was selected based on the coefficient of variation of internal metrics and Landsat data. Ambient wind conditions were characterized using raster data from the Global Wind Atlas (GWA, 2022), reporting mean monthly wind speeds at 10 m above ground ranging from 3.7 m/s at lower elevations to 9.4 m/s at upper elevations over 2015–2020. Mean annual precipitation of 596 mm (1981–2010) was obtained from the PRISM Climate Group (Daly and Bryant, 2025).

Descriptive statistics for all covariates, including the disturbance classes described in Sections 2.3 and 2.4, were calculated by pooling observations and applying ratio estimation at each location (Wiegand and Moloney, 2013) (Tables S1 and S2).

2.2. Image processing

Multispectral satellite images were acquired from the Landsat 7 ETM+ and Landsat 8 OLI archives (Table 2). Pre- and post-disturbance scenes were selected from cloud-free days (<20% cloud cover) between 1 June and 30 August. Images underwent Level 1 terrain correction and Level 2 atmospheric correction. Georeferencing was performed using ground control points and a digital elevation model to improve topographic accuracy. Radiometric and aerosol corrections followed Landsat surface reflectance protocols (Martin et al., 2016; Mobley, 2015).

Beetle outbreak mapping drew on a time series of inter-annual vegetation-differenced images from Landsat 7 ETM+ and Landsat 8 OLI spanning nine years (2005–2014). Wildfire burn severity mapping used paired pre- and post-disturbance Landsat collections to derive vegetation-differenced indices for classification. Table 2 lists all raster collections used in both analyses.

2.3. Wildfire mapping

Reference data for burn severity classification were generated from 94 burn plots: 9 field plots established immediately post-fire in 2015 and resampled in 2020, and 85 aerial photo-interpreted plots. All plots were scored using a composite burn index (CBI) framework adapted from Key and Benson (2005); Hope et al., (2016); Schepers et al., (2014). Aerial plot scoring followed the methods of Collins et al. (2018). CBI values were assigned as weighted estimates of fire effects across four vegetation layers: substrate, ground shrubs, understory, and overstorey. Severity was ranked on a scale of 0.00–10.00, where < 2.50 indicated no change, 2.50–5.00 low burn, 5.00–7.50 moderate burn, and > 7.50 high burn (Table 3).

Burn plots were coded as digitally processed training samples for classification of five spectral indices in ArcMap v.10.7. Candidate indices were selected based on demonstrated predictive power in published wildfire studies (Table 4). Plot locations were mapped prior to field survey, targeting the center points of areas with relatively uniform disturbance conditions within a four-cell Landsat pixel quadrant (60 m x 60 m). Where burn severity classes were intermixed, a minimum plot diameter of 20 m and minimum inter-class spacing of 45 m were enforced to mitigate spatial autocorrelation and ensure composite spectral signatures for each class (Lentile et al., 2006; Trigg and Flasse, 2001).

The sensitivity of spectral variables to burn effects was assessed in three stages: separability index analysis, analysis of variance, and regression analysis. The separability index was computed as:

$$M = \frac{\|u_2 - u_1\|}{\sigma_2 + \sigma_1}$$

where μ_1 and μ_2 are the mean spectral values of the target burn class and

Table 2

Landsat satellite imagery acquired for beetle outbreak mapping and wildfire burn severity classification, with sensor, acquisition date, and spatial resolution.

Scene	Satellite sensor	Date of acquisition	Resolution (m)
Pre-outbreak	Landsat 5	2003 Aug 20	< 30 m (~0.0088°)
Early-outbreak	Landsat 5	2005 Aug 09	< 30 m (~0.0088°)
Early-outbreak	Landsat 5	2006 Aug 28	< 30 m (~0.0088°)
Early-outbreak	Landsat 5	2007 Aug 15	< 30 m (~0.0088°)
Mid-outbreak	Landsat 5	2008 Aug 17	< 30 m (~0.0088°)
Mid-outbreak	Landsat 5	2009 Aug 20	< 30 m (~0.0088°)
Mid-outbreak	Landsat 5	2010 Aug 07	< 30 m (~0.0088°)
Post-outbreak	Landsat 5	2011 Aug 10	< 30 m (~0.0088°)
Post-outbreak	Landsat 5	2013 Aug 31	< 30 m (~0.0088°)
Pre-fire (–1yr)	Landsat 8	2014 July 01	< 30 m (~0.0088°)
Pre-fire (–1mth)	Landsat 8	2015 July 04	< 30 m (~0.0088°)
Post-fire (+1 yr)	Landsat 8	2016 July 20	< 30 m (~0.0088°)
Post-fire (+5 yrs)	Landsat 8	2020 Sept 03	< 30 m (~0.0088°)

Table 3

Composite burn index scoring criteria used to classify wildfire burn severity across four vegetation layers, adapted from Key and Benson (2005).

Vegetation layer	Burn severity scale				
	Weighting	Unburned	Low	Medium	High
Substrates					
Charred litter & snag (%)	1.00	0.00%	< 50%	50%–90%	> 90%
Charred duff & fine roots (%)	1.00	0.00%	< 25%	25%–50%	> 75%
Ground shrub (<1 m)					
Charred foliage (%)	1.00	0.00%	< 25%	25%–50%	> 75%
Resprouting (%)	1.00	100%	> 90%	30%–90%	< 30%
Understory (1–5 m)					
Charred foliage (%)	1.00	0.00%	< 10%	10%–50%	> 50%
Residual/resprouting (%)	1.00	100%	> 90%	20%–90%	< 20%
Overstorey (>5 m)					
Charred height (m)	1.00	0.00 m	< 1.8 m	1.8–4 m	> 4 m
Canopy black (%)	2.00	0.00%	< 20%	20%–80%	> 80%
Canopy green (%)	1.00	100%	> 80%	20%–80%	< 20%

Table 4

Candidate spectral indices evaluated for wildfire burn severity classification, with formulas and source references.

Spectral index	Formula	References
Normalized Burn Ratio	$\frac{(NIR - SWIR2)}{(NIR + SWIR2)}$	(Key and Benson, 2005)
Burned Area Index 2	$\left(1 - \sqrt{\frac{NIR}{RED}}\right) \times \left(\frac{SWIR2 - NIR}{\sqrt{SWIR2 + NIR}} + 1\right)$	(Filipponi, 2018)
Mid-InfraRed Burn index	$10 \times SWIR2 + 9.8 \times SWIR1 + 2$	(Tran et al., 2018; Trigg and Flasse, 2001)
Thermal Veg Index	$\frac{(NIR - TIR)}{(NIR + TIR)}$	(Smith et al., 2007)
Thermal Veg Index ^{SWIR}	$\frac{(NIR - SWIR2 \times TIR1)}{(NIR + SWIR2 \times TIR1)}$	(Holden et al., 2005)

pooled remaining classes, and σ_1 and σ_2 are the corresponding standard deviations (Schepers et al., 2014; Tran et al., 2018). Given non-normal distributions in spectral variables, non-parametric tests were used: the Kruskal-Wallis H-test and Wilcoxon rank-sum test.

Separability analysis of individual spectral bands and composite indices (Tables S3, S4) confirmed that NBR provided the strongest overall discrimination between burn severity classes ($M = 0.399$, $p < 0.001$), producing the highest values ($M = 1.484$). For moderate burn and unburned classes, only the NIR band yielded significant separability ($M = 0.748$, $p < 0.001$).

MIRBI showed significant discriminatory power for high and low burn severity ($M > 0.740$, $p < 0.001$) but was ineffective for moderate burn and overall burned-versus-unburned detection ($M < 0.016$). NBR was significant across both overall ($p < 0.001$) and individual burn class distinctions ($p < 0.05$), making it the strongest candidate for wildfire mapping.

Regression diagnostics assessed predictive relationships between spectral indices and CBI values using support vector machine (SVM) kernels with optimized cost margins of 0.316–1.105 (Table 5). Covariance (R^2), prediction accuracy (RMSE), and overfitting risk (RMSE ratio) were evaluated via repeated 10-fold cross-validation. NBR showed the highest correlation with field-measured CBI values ($R^2 = 0.879$) and the



Fig. 1. A, B, C. Three high severity burn plots within the Mt Midgeley Wildfire site, with composite burn index scores of 7.91, 8.66, and 7.56 (left to right). Standing dead snags, deep charring, and downed coarse debris characterize the overstory, 5-years post-fire.

Table 5

Support vector machine diagnostics for five indices predicting composite burn index values. Model performance assessed by coefficient of determination (R^2), root mean square error (RMSE), overfitting ratio ($RMSE^{ratio}$, training/validation), and systematic bias (Theil's U^{bias}). Hyperparameters epsilon (ϵ) and cost (C) control noise tolerance and penalty, respectively (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

	R^2	RMSE	$RMSE^{ratio}$	U^{bias}	C	ϵ
NBR	0.879***	1.05	0.46	0.035	0.316	0.10
BAI2-SL	0.183***	2.61	1.09	0.307	1.105	0.10
MIRBI	0.551***	1.94	1.16	0.469	0.316	0.10
TVI	0.584***	1.88	1.16	0.432	0.474	0.10
TVI-SW	0.241***	2.54	1.10	0.343	0.158	0.10

lowest prediction error ($RMSE = 1.05$), reducing error by more than 0.8 CBI units compared with TVI ($RMSE = 1.88$) and MIRBI ($RMSE = 1.94$). TVI and MIRBI were rejected due to overfitting metrics exceeding the 1.1 $RMSE^{ratio}$ threshold. Theil's U-statistic confirmed overall bias above recommended levels for both ($U^{bias} > 10\%$).

Based on sensitivity analyses and regression diagnostics, NBR was selected as the optimal predictor of wildfire burn severity. To confirm classification accuracy, the dataset was partitioned into 75% training and 25% validation subsets. A support vector machine classifier was trained and evaluated using 10-fold cross-validation. Confusion matrices (Table 6) confirmed NBR as the top performer ($\kappa = 0.792$, $OA = 0.872$). However, NBR exhibited notable single-class misclassification errors for low burn intensity (commission > 0.216 , omission < 0.554). The NBR index was applied to classify a vegetation-differenced image derived from pre- and post-disturbance rasters (2014–2016) (Fig. 2). The resulting classified burn map was compared with wildfire reporting data from the BC Wildfire Service for the Mt Midgeley fire.

Pixels in the unburned class correspond to $dNBR < 0.10$ and a CBI of < 2.50 in matched reference plots (Key and Benson, 2005), representing within-perimeter areas where canopy vegetation showed no detectable mortality. These include both truly unburned refugia and locations of low-intensity surface fire that left canopy foliage intact; the two were not separated because the regeneration-relevant outcome — preserved canopy seed supply for fire-evaders and fire-resisters — is the same in both.

Table 6

Classification accuracy of candidate spectral indices for discriminating wildfire burn severity classes (high, moderate, low, unburned). Overall accuracy (OA) and per-class kappa coefficients (κ) reported for each index. Ground-truth classes derived from composite burn index field plots.

	Commission error					Omission errors				
	Kappa	HB	MB	LB	UB	HB	MB	LB	UB	OA
NBR	0.79	0.02	0.00	0.22	0.20	0.04	0.00	0.55	0.02	0.87
BAI2-SL	0.62	0.00	0.06	0.28	0.52	0.30	0.00	0.00	0.00	0.73
MIRB	0.60	0.00	0.00	0.48	0.13	0.00	0.15	0.19	0.00	0.70
TVI	0.31	0.72	0.00	0.66	0.21	0.63	0.11	0.02	0.25	0.46
TVI-SW	0.66	0.18	0.05	0.34	0.51	0.34	0.07	0.37	0.03	0.50

2.4. Beetle mapping

Red-stage beetle outbreak was mapped as an indicator of MPB-caused tree mortality using inter-annual Landsat imagery from 2005 to 2014 and four spectral indices selected from published sources (Table S5) (Collins and Woodcock, 1996; Jin and Sader, 2005; Walter and Platt, 2013; Woodward et al., 2018). Reference data were sampled from 28 square field plots (20×20 m) established in beetle-affected areas during July and August 2020. Applying sampling protocols of Carlson et al. (2017), plot locations were mapped in ArcMap 10.7, targeting diverse topographic settings and outbreak severities, with plot centres geopositioned to align with designated Landsat pixel quadrants (60×60 m). To confirm that red-attack evidence was attributable to *Dendroctonus ponderosae*, tree boles were surveyed for pitch tubes, frass texture and colour, and egg gallery architecture (Amman, 1982; Hagle et al., 2003; Lerch et al., 2016). Outbreak intensity was quantified as the fraction of basal area composed of beetle-killed pine relative to total plot basal area (400 m^2), with plots retained for model training only if red-stage mortality exceeded 80%.

Spectral sensitivity to beetle damage was assessed through separability analysis, analysis of variance, and linear regression (Tables S6, S7). NDMI emerged as the top performer, with the highest separability score ($M = 1.004$) and strong rank-sum test significance ($p = 0.00000008$). discrimination of red-stage outbreaks ($M = 1.135$). Aerosol, blue, long-shortwave infrared, and red bands showed moderate discriminatory power ($M < 0.3$), though long-shortwave and red bands outperformed others in variance testing ($p < 0.000004$).

Regression diagnostics used SVM kernels with optimized cost margins of 0.316–2.368 (Table 7). NDMI showed a strong correlation with the basal area fraction of beetle-killed trees ($R^2 = 0.905$) and the lowest error ($RMSE = 4.83$) without overfitting.

Image classification proceeded in two stages: initial mean segmentation with unsupervised classification (ISODATA), followed by supervised classification using a support vector machine. Reference data were stratified proportionately by class and partitioned into 75% training and 25% validation subsets. Confusion matrices (Table 8) confirmed NDMI as the top performer ($\kappa = 0.750$, $OA = 0.824$), with acceptable error rates across all single-class classifications (0.133–0.199). NDMI was applied to classify red-stage outbreak from a series of vegetation-differenced images derived by subtracting the 2005 pre-disturbance scene from each annual outbreak scene (Fig. 3).

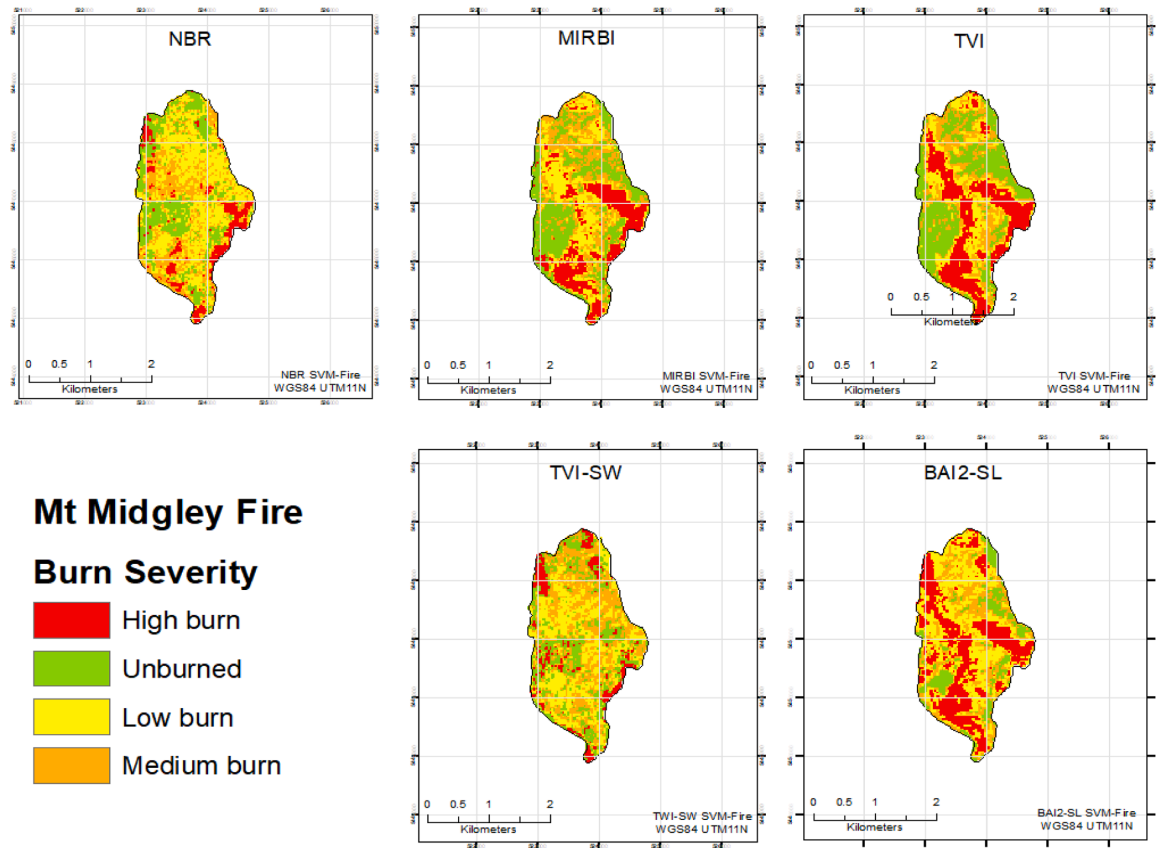


Fig. 2. Results of classification of wildfire burn severity based on five spectral indices.

Table 7

Regression diagnostics comparing performances of spectral variables to predict beetle outbreak severity based on levels of covariance (R^2); root mean square error (RMSE); risk of overfitting ($RMSE^{ratio}$); and model bias (U^{bias}). Hyperparameters fitted for features of noise tolerance or insensitivity loss function as Epsilon (ϵ) and model Cost value or measure of the penalizing sigma (C) (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

	a(B)	R^2	RMSE	$RMSE^{ratio}$	U^{bias}	Cost-C	Gamma
NDMI	0.738 (0.863)	0.905***	4.83	1.09	0.176	0.947	0.10
TAS-Wet	2.131 (0.804)	0.904***	4.92	1.69	0.049	0.316	0.10
TAS-Green	3.250 (0.764)	0.650***	8.87	1.16	0.030	2.368	0.10
TAS-Bright	3.523 (0.738)	0.679***	8.36	1.16	0.032	1.263	0.10

Table 8

Accuracy assessment of spectral predictors to detect areas of beetle outbreak. Results reported on overall accuracy assessment (OA) and according to kappa scores of classifications of red attack stage tree mortality (RA) and elsewhere (Non-RA).

	Commission Errors			Omission Errors		OA
	Kappa	RA	Non-RA	RA	Non-RA	
NDMI	0.750	0.198	0.133	0.199	0.166	0.824
TAS-wet	0.693	0.158	0.271	0.158	0.242	0.796
TAS-bright	0.605	0.250	0.424	0.211	0.416	0.730
TAS-green	0.570	0.290	0.625	0.380	0.502	0.861

Recognized as the optimal index, NDMI was chosen to predict red-stage outbreak from a series of vegetation-differenced images derived by subtracting the 2005 pre-disturbance scene from the annual outbreak scenes (Fig. 3).

Grey-stage beetle outbreak at the time of the 2015 Mt Midgley fire was derived as the union of the annual red-attack NDMI classifications

for 2005 through 2011 inclusive. Each annual raster Beetle.Outbreak.YYYY.tif is a Landsat scan-year classification of pixels with a red-attack NDMI signature in that year's imagery, and individual pixels may register in more than one annual scene while needles remain red. A pixel was classified as grey-stage if it carried a red-attack classification in any year between 2005 and 2011, on the rationale that at least four years have then elapsed by 2015 for needles to fall and crowns to collapse, satisfying the operational grey-stage definition (outbreak 3–15 y before fire) of Carlson et al. (2017) and the foliage-fall and snag-collapse chronologies of Hicke et al. (2012) and Meddens et al. (2012). The 2012 annual raster is absent from the project archive; a pixel attacked solely in 2012 and not re-registered in either 2011 or 2013 is therefore not captured, although in practice persistent red signal in adjacent years carries forward most 2012-affected pixels. Detections in 2013 and 2014 were retained as red-stage only, since needle fall is incomplete within two years of attack. The composite was stored as mpb_grey_na.tif at the 30 m native grid of the source rasters and entered as a binary covariate (1 = grey, 0 = not grey) in the inhomogeneous log-Gaussian Cox process model. Region-wide the raster contains 1541 grey-stage cells; within the

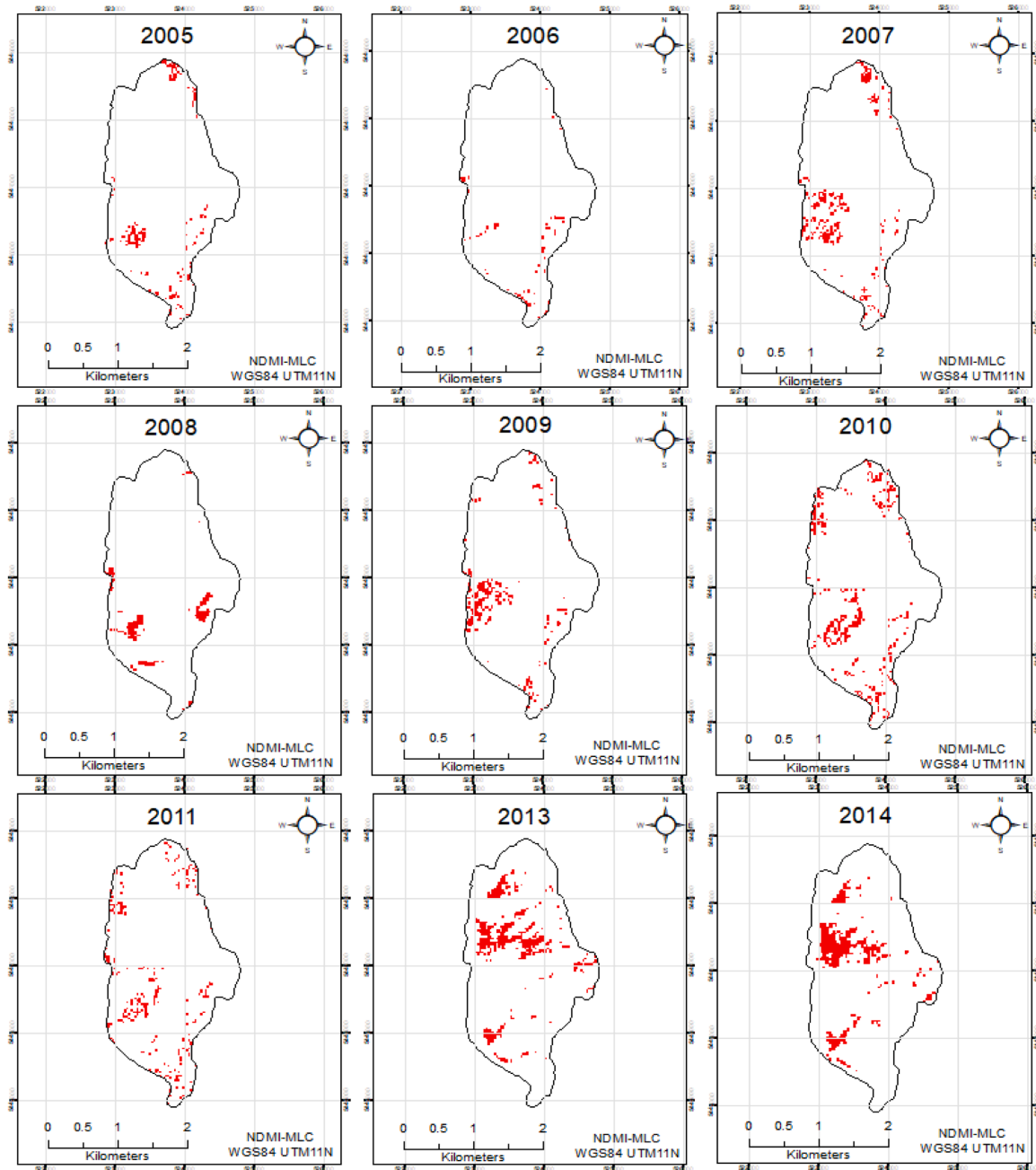


Fig. 3. Results of classification of red stage tree mortality (2005–2014).

three 4-ha plot footprints it contains 18, 47, and 3 cells for Plots 1, 2, and 3 respectively (68 pooled, calculated with all-touched edge inclusion at the 30 m grid). Those cells host 274, 949, and 92 of the 3912 mapped seedlings, 1315 pooled, 33.6% of the analysed set, which is the unit of observation on which the Cox process model is fitted.

2.5. Regeneration models

Throughout this paper, seedling intensity refers to the first-order property $\lambda(s)$ of the inhomogeneous log-Gaussian Cox process: the expected number of seedlings per unit area at location s (stems m^{-2}). This usage is standard in spatial point-pattern analysis (Baddeley et al., 2015;

Diggle, 2013; Illian et al., 2008) and in published applications to post-disturbance conifer regeneration (Chambers et al., 2016; Malone et al., 2018; Owen et al., 2017; Wild et al., 2014; Ziegler et al., 2017). The response variable is the intensity function $\lambda(s)$ of an inhomogeneous log-Gaussian Cox process: the expected number of seedlings per unit area at location s , evaluated continuously across each plot. Covariate values are extracted at seedling locations from the 30-m covariate stack; intensity is estimated on a 5-m integration mesh using the spatstat R package (Baddeley et al., 2015). This formulation preserves the sub-metre resolution of stem-mapped seedling locations while accommodating the coarser-grained covariate fields.

Point process models were used to investigate the spatial patterns of

conifer regeneration in relation to seed source proximity, topography, wind, and MPB and fire disturbance histories. The analysis proceeded in two stages. First, simulations of complete spatial randomness were tested against known seedling locations. Second, Cox process regression models fitted with a distance-weighted log-Gaussian function were used to estimate the significance and direction of covariate effects on seedling intensity (stems m^{-1}). All models were governed by parsimony, with predictor screening aimed at minimising environmental collinearity and addressing spatial autocorrelation among clusters (Agovino et al., 2021; Klippel et al., 2011; Tapia-McClung, 2006).

Conifer seedlings were sampled in summer 2020 across three 4-ha plots (200 m $\times$ 200 m) located within the Mt Midgeley fire perimeter. Plots were purposively distributed across gradients of fire severity and beetle-outbreak history, and each was positioned within 200 m of residual forest to capture the effect of seed source proximity on regeneration. Within every plot every conifer seedling was recorded by sub-metre GPS as a stem-mapped point, along with species, height (cm), the azimuth and distance to the nearest reproductive parent of the same species (the parent's DBH was also recorded), and a burn-site categorical call (Low / mid-High) made in-field against the dNBR class boundaries described in §2.3. Observed seedling heights ranged from 0 to 60 cm (median 15 cm); only individuals consistent with post-2015 establishment were retained, judged in-field by applying species-specific annual height increments from regional silvicultural growth records (Agee, 1996) to a five-year (2015–2020) post-fire window. Where several conspecific seedlings occurred in close proximity to one another (within a small radius, typically below 1 m), a greyingle central GPS point was recorded for the cluster with a stem-count attribute and a mean-height attribute, in line with the stem-mapping protocol; the operative threshold and a sensitivity check on cluster reconstruction are given in Supplementary S3. Species recorded with fewer than 30 individuals across all three plots were excluded. In total, 3912 seedlings of seven conifer species were retained for analysis (plot 1, $n = 1037$; plot 2, $n = 1268$; plot 3, $n = 1607$).

Within each 200 m $\times$ 200 m plot, every post-fire conifer seedling was stem-mapped from a single central anchor using a Nikon Forestry Pro hand-held laser rangefinder for distance to stem and a Suunto MC-2 compass for azimuth, with measurements logged to a GeoODK form on a field tablet. The plot anchor was georeferenced with a handheld GPS receiver in EPSG:3153, anchoring the local stem-map grid to global coordinates; polar measurements were post-processed to x,y positions in that projection. Distance from each seedling to the nearest cone-bearing conspecific seed source was measured directly with the same rangefinder. Across the 3912 mapped seedlings, seed-source distances ranged from 1.57 to 169.26 m (median 27.68 m).

All disturbance and covariate rasters were aligned to a common 30 m grid in EPSG:3153, the native resolution of the Landsat-derived dNBR and dNDMI products. Terrain layers derived from the 1 m HRDEM (elevation, slope, aspect, TWI, RIX) were resampled from their processed 25.11 m intermediate grid to 30 m using bilinear interpolation; the Global Wind Atlas product (native 25.09 m) was resampled identically. Categorical layers (burn severity class, MPB stage) were resampled with nearest-neighbour. Seedling locations were retained as continuous sub-metre point data; covariate values were extracted at each seedling location from the 30 m grid rather than aggregated to it, preserving the stem-mapped resolution of the response variable.

Non-spatial summary measures, including absolute density (ha^{-1}), nearest-neighbour distance, and distance to seed source, were calculated by pooling observations and applying ratio estimation (Table 9) (Wiegand and Moloney, 2013). Spatial dependence was assessed using Ripley's K-function, with observed seedling intensity compared against a null model of non-constant intensity through 999 Monte Carlo simulations (Larson and Churchill, 2012; Malone et al., 2018). The null model was fitted with a nonparametric Epanechnikov kernel (Baddeley et al., 2015; Chambers et al., 2016; Wiegand and Moloney, 2013; Ziegler et al., 2017) and an isotropic edge correction (Yamada and Rogerson, 2003).

Table 9

Likelihood ratio test of covariate importance.

Covariate	β	Deviance	LR-test
SS-dist	−0.0125	1065.27	0.001***
Elevation	−0.0004	29.88	0.001**
Slope	−0.0062	7.25	0.007**
Aspect	0.3622	136.51	0.001***
RIX	1.3563	20.32	0.001***
TWI	0.0421	15.29	0.001***
Wind	−0.0662	9.57	0.002**
MPB-grey	−0.0233	0.47	0.492
MPB-red	−0.1461	14.85	0.001***
Fire-high	−0.6430	211.64	0.001***
Fire-med	0.0188	0.34	0.563
Fire-low	0.5682	276.87	0.001***

Goodness-of-fit tests assessed the strength of spatial clustering across distances of 0–15 m. Cross-validation identified an optimal bandwidth of 11 m (Pielou, 1962; Van Es, 1991). Bivariate pair correlation functions were then fitted to examine interspecific spatial associations at varying distances (Chambers et al., 2016; Ziegler et al., 2017).

To quantify the effects of covariates on seedling intensity, an inhomogeneous, locally weighted Cox process model was fitted. Candidate covariates were screened using likelihood ratio tests, comparing the linear slope estimate for each predictor against an intercept-only model and excluding those that did not significantly improve model deviance ($p < 0.05$; Table 9) (Meddens et al., 2013, 2012). Retained predictors were then evaluated using penalized LASSO regression, in which a tuning constraint minimized the residual sum of squares and scored covariates by the persistence of their coefficient magnitude (Fonti and Belitser, 2017; Larson and Churchill, 2012). This yielded a final set of eight covariates: seed source distance, aspect, terrain ruggedness, wind, MPB grey-stage, MPB red-stage, high-severity fire, and low-severity fire.

The final model was specified as an inhomogeneous log-Gaussian Cox process. This formulation estimates the probability of seedling intensity at a given location as a function of both measured covariates and unmeasured spatial heterogeneity, the latter captured by a latent Gaussian random field with zero mean and variance-covariance matrix Σ of dimension $q \times q$, weighted by distance from the observation (Illian et al., 2012; Renner et al., 2015). For seedling outcome y^{ij} of the i th observation at the j th location, intensity was modelled as a function of fixed covariate effects and random effects u^{ij} representing residual spatial variation. The expected intensity Y^i of species i was fitted as:

$$Y_i = \exp(\beta_1 SSdist_i + \beta_2 RA_i + \beta_3 GA_i + \beta_4 BS_i + \beta_5 RIX_i + \beta_6 AS_i + \beta_7 W_i + \epsilon_i)$$

where SSdist is distance to seed source, RA and GA are indicators of red-stage and grey-stage beetle mortality, BS is burn severity class, RIX is terrain ruggedness, AS is aspect, W is mean post-fire wind speed, and ϵ_i is the residual error term.

3. Results

3.1. Spatial patterns of conifer regeneration density between plots

We sampled 3886 post-fire seedlings across three plots. Mean regeneration density (ha^{-1}) significantly differed between plots ($h = 342.97$; $df = 2$, $p < 0.000$), with higher density in plot 1 ($318.0 \pm 122.2ha^{-1}$) than plot 2 ($230.4 \pm 103.4ha^{-1}$) and plot 3 ($306.6 \pm 140.2ha^{-1}$, Table 10). Plot location significantly influenced the mean distance to nearest neighbour ($h = 440.03$, $df = 2$, $p < 0.000$) and distance to nearest seed source ($h = 683.68$, $df = 2$, $p < 0.000$). In plot 1, seedlings were clustered closer to each other (1.82 m) and to their nearest seed source (20.68 ± 17.57 m) than in other plots. Spatial patterns of seedling intensity (m^{-2}) varied by plot Fig. 5). In plot 1, dispersion was significant beyond 9 m, while in plot 3, significant

Table 10

Mean values (Stdev) reporting absolute 'Density', nearest-neighbour distance 'NNdist', and distance to seed source 'SSdist' of all conifer seedlings. Significance of variance in density and nearest neighbour distance reported at 0.05, 0.01, 0.001 levels as *, **, *** according to Kruskal Wallis test. Deviations from complete randomness towards clustering (<1.000) and dispersion (>1.000) reported at 0.05, 0.01, 0.001 levels as #, ##, ### according to Clark & Evans Aggregation test.

	Density (ha ⁻¹)	NNdist (m)	SSdist (m)	CE-index
Pooled	305.5	1.73 (—)	79.0 (—)	0.177###
Plot 1	245.7	0.92 (2.19)	44.0 (51.5)	0.583###
Plot 2	295.9	2.44 (1.91)	79.0 (39.0)	1.005
Plot 3	370.0	1.73 (3.26)	91.0 (58.0)	0.807###

patterns extended up to 24 m and 22 m, respectively.

These spatial trends were contrary to descriptive analysis of plot disturbances (Table S2, Fig. 4). In plot 1, with dispersed seedlings, larger areas experienced high severity burn ($h = 15.83$, $df = 2$, $p = 0.0004$) and moderate severity burn ($h = 137.19$, $df = 2$, $p < 0.000$). Meanwhile, in plot 2, where seedlings clustered, significantly greater areas were affected by red-stage outbreak ($h = 969.49$, $df = 2$, $p < 0.000$). (Fig. 5)

3.2. Spatial patterns of post-fire regeneration density between species

Across the total sample, there was statistically significant variation in mean seedling density between species ($h = 1568$, $df = 7$, $p < 0.000$) (Table 11). Lodgepole pine exhibited the highest relative density, comprising 73.5% of the post-fire seedlings sampled. Subsequent species with discernible contributions were Western larch (8.2%) and Engelmann spruce (7.4%).

Spatial analyses revealed distinct clustering patterns among all conifers ($p < 0.05$). Substantial deviations from complete spatial randomness were identified among Lodgepole pine, Douglas-fir, Engelmann spruce, and Western hemlock at intermediate scales ranging from 10 to 55 m. Additionally, significant departures from complete spatial randomness (CSR) were noted among Western larch, Ponderosa pine, and Western white pine at distances spanning from 60 to 180 m (Fig. 6).

3.3. Spatial patterns of seedling intensity in response to covariates by species

Cox process models indicated that seedling density (the intensity $\lambda(s)$) of all conifer seedlings combined were positively associated with low severity fire ($p < 0.001$), while increasing distance to seed sources, higher wind speeds, red-stage beetle outbreaks, and high severity fire were negatively associated ($p < 0.01$, Table 12). Specifically, in areas with low severity fire, seedling intensity among conifers collectively increased by 18.1%. Conversely, intensity decreased by 18.9% in high severity fire areas and by 14.3% in regions affected by red-stage beetle outbreaks

When disaggregated by species, high severity fire had significantly adverse effects on regeneration patterns for four species: Lodgepole pine, Douglas-fir, Ponderosa pine, and Western hemlock ($p < 0.05$). The strength of these relationships varied markedly. Notably, in high burn areas, seedling intensity decreased most significantly for 'fire-avoiders' like Western hemlock ($\beta = -1.851$), followed by 'fire resisters' like Ponderosa pine ($\beta = -1.030$) and Douglas-fir ($\beta = -0.627$). In contrast, Lodgepole pine exhibited a comparatively modest decrease of 17.3%, in part confirming our first hypothesis (H1).

Combined disturbances from beetle outbreaks and wildfires significantly affected seedling patterns for only one species, Lodgepole pine. Additionally, the effects of beetle outbreaks, though variable between host and non-host species, yielded unexpected outcomes. Red-stage beetle outbreaks negatively impacted seedling intensity of two host species, Lodgepole pine ($\beta = -0.167$, $p < 0.001$) and Western white pine ($\beta = -1.151$, $p < 0.05$). Conversely, grey-stage beetle outbreaks

had no significant effect on seedling intensity of any species.

Wind exerted significantly adverse effects on the regeneration of four conifer species, including Lodgepole pine, Western larch, Engelmann spruce, and Ponderosa pine ($p < 0.05$). In contrast, for Western white pine, wind demonstrated a significant positive impact on patterns of seedling intensity ($p < 0.01$). Specifically, in areas where the mean wind speed exceeded the site average by 1 m/s (4.9 m/s; see Table S1), the seedling intensity of Western white pine increased by a substantial 146.9%. Furthermore, Western white pine exhibited greater abundance on southern-facing aspects ($p < 0.001$), distinguishing it from other conifers that thrived on eastern and north-eastern aspects.

4. Discussion

In this study, we examined the spatial patterns of post-fire conifer regeneration following disturbance events of mixed-severity wildfire and mountain pine beetle outbreaks. Our exploration centred on three hypotheses: (H1) anticipating lower seedling intensity in high severity fire zones for different conifer groups; (H2) expecting diminished seedling intensity in areas of beetle outbreak for host species compared to non-hosts; and (H3) projecting reduced seedling intensity in locations experiencing combined disturbances from both high severity fire and beetle outbreaks for host species versus non-host species.

Our results offered partial support for these hypotheses. 'Fire evaders,' notably Lodgepole pine, exhibited significantly higher seedling intensity in high severity fire areas compared to 'fire resisters' such as Douglas-fir and Ponderosa pine. Additionally, seedling intensity was notably lower among host species (Lodgepole pine and Ponderosa pine) than among non-host species in areas of beetle outbreak, aligning with our expectations. However, the effects of combined disturbances were only significant for Lodgepole pine and negligible for Western white pine and Ponderosa pine. Intriguingly, our models highlighted the pivotal role of abiotic factors, specifically topography and wind, in influencing the spatial patterns of post-fire and post-beetle conifer regeneration among both host and non-host species.

4.1. Species responses to mixed severity fire events

The findings above regarding plot-level and global variance in seedling density present conflicting evidence concerning our primary hypothesis H1 and secondary hypothesis H2. On one hand, the spatial distribution of seedling density implies a detrimental impact of wildfire on regeneration across plots, as dispersion patterns align with larger areas affected by higher severity burn. Conversely, variations in seedling density (ha⁻¹) between plots indicate a positive influence of wildfire on regeneration patterns, with the highest density observed in plot 1. Accordingly, it becomes imperative to apply finer scale analysis between species and within plots to better understand seedling patterns and disturbance interactions.

This suggests a pivotal role of topography and wind in the dispersion and establishment of specific conifers under changing environments. Considering the biogeoclimatic history of Western white pine in the Creston Valley, it is plausible that average wind vectors from a westerly direction favour its downhill expansion as a 'fire invader' into exposed areas subsequent to the wildfire disturbance.

Historically, wildfire and beetle outbreaks have been pivotal in shaping the dynamics of forest ecosystems in Southern Interior British Columbia. Existing literature underscores the diverse regeneration strategies of conifers as key determinants in stand competition between early- and late-seral species (Agee, 1998; Glenn-Lewin et al., 1992; McIntosh, 1980; Pickett and White, 1987). Our findings align with these theoretical frameworks, revealing a nuanced relationship between post-fire regeneration and varying traits of post-fire seed dispersal and fire-resilience. Notably, wildfire had more pronounced effects on 'fire avoiders' and 'fire resisters,' such as Western hemlock, Douglas-fir, and Ponderosa pine, which possess characteristics making them susceptible

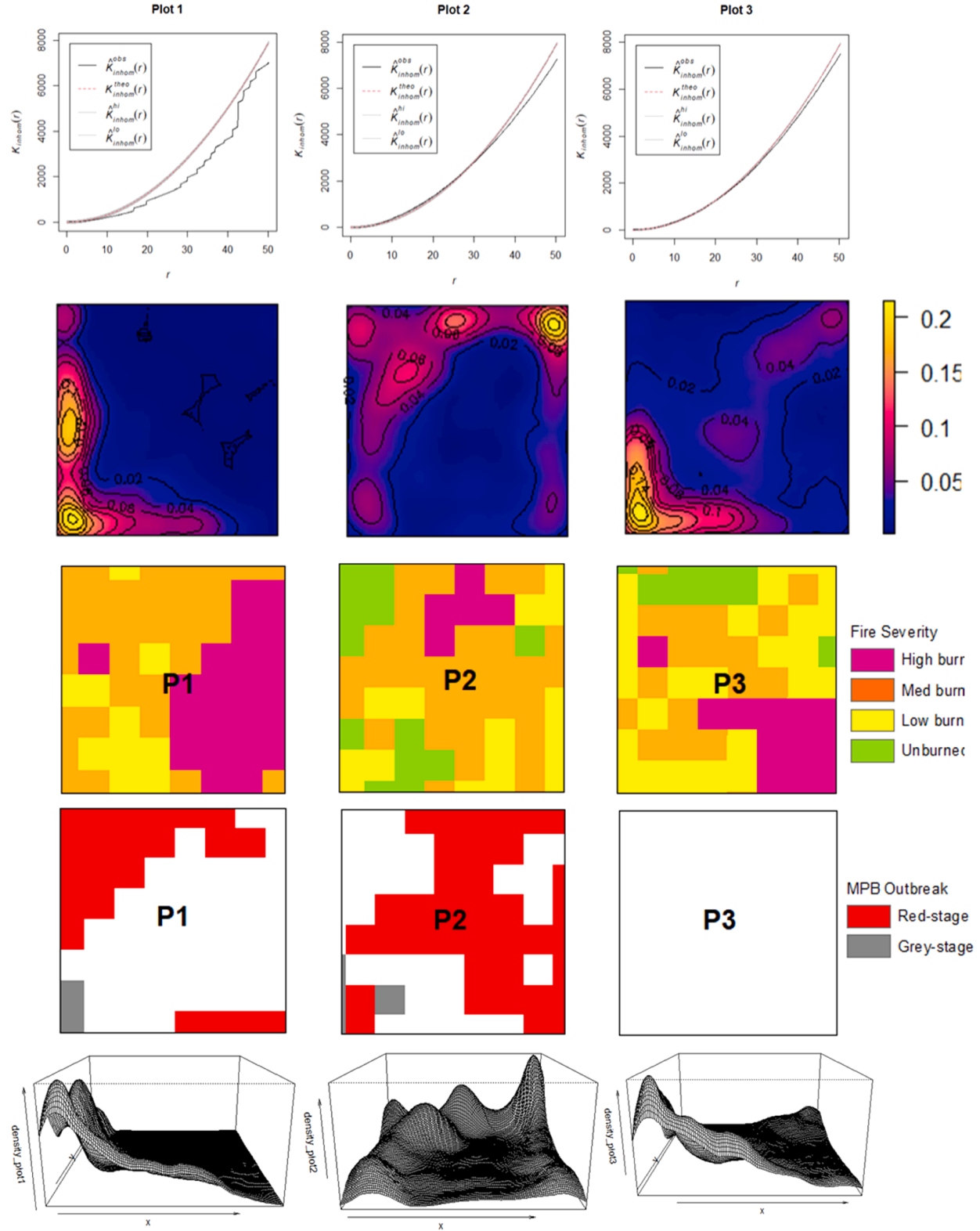


Fig. 5. Top row: Ripley's K-diagrams showing regeneration patterns of dispersion and clustering below and above the null model in each plot. Second from top row: Kernel density heat maps depicting seedling intensity at a bandwidth of 10 m. Third from top row: Diagram of stem mapped data by species. Bottom row: Perspective maps representing kernel density on a 3-dimensional plane.

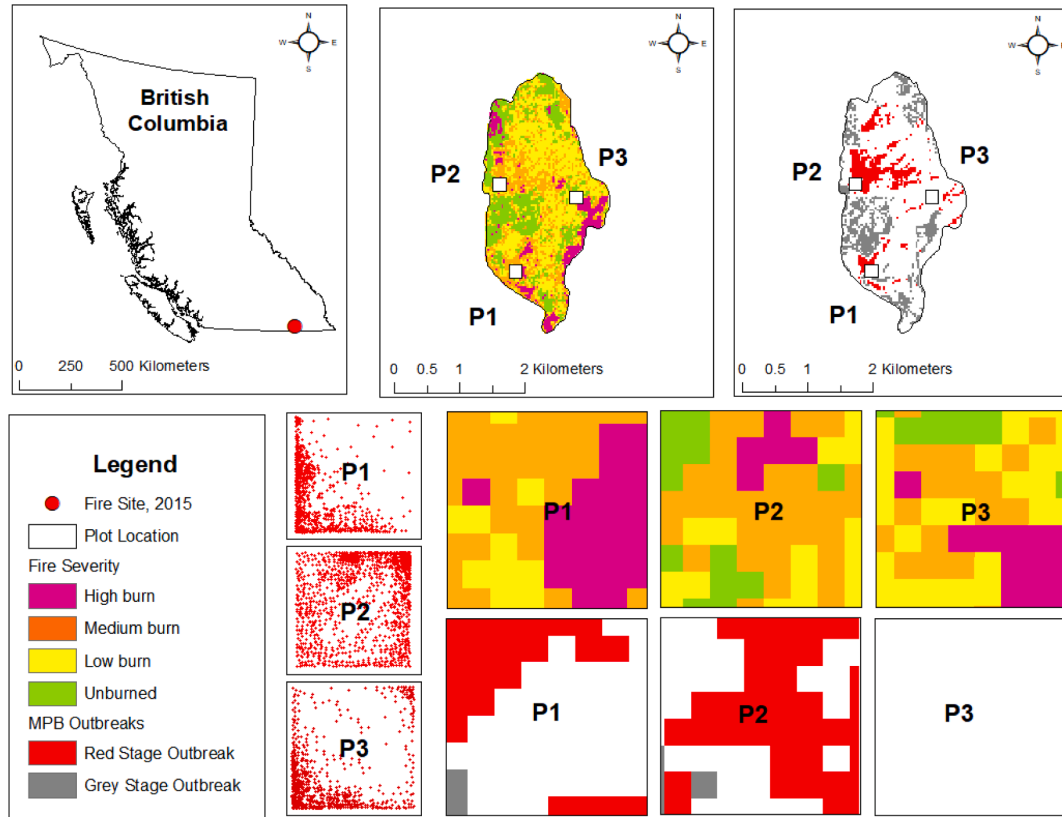


Fig. 4. Classified multispectral Landsat imagery of the 2015 Mt Midgeley Fire footprint. Top centre: dNBR burn severity classification. Top right: dNDMI red-stage MPB classification, 2005–2014. Each of the three study plots (P1, P2, P3) is a 200 m × 200 m (4 ha) square. Red dots within each plot panel mark individual mapped seedling locations ($n = 3912$ total).

Table 11

Mean values (Stdev) reporting absolute ‘Density’, nearest-neighbour distance ‘NNdist’, and distance to seed source ‘SSdist’ seedlings by species across all plots. *Kruskal Wallis nonparametric test of variance: values reported in bold significantly higher than other plots at 0.05, 0.01, 0.001 levels as *, **, ***. Significant deviations from CSR towards clustering (<1.000) and dispersion (>1.000) shown in bold at 0.05, 0.01, 0.001 levels as #, ##, ### according to the Hopskel and Skellam index.

	Density _{ha} –1 (%)	NNdist (m)	SSdist (m)	HS-index
<i>Pinus contorta</i>	226.7 (74.4%)	2.14 (2.79)	78.0 (58.0)	0.803**
<i>Larix o.</i>	25.1 (8.3%)	6.17 (6.51)	71.0 (60.0)	0.844**
<i>Picea e.</i>	22.9 (7.5%)	6.63 (6.45)	77.0 (52.0)	0.833**
<i>Pseudo. M.</i>	16.5 (5.4%)	8.16 (8.31)	76.0 (45.0)	0.843**
<i>Pinus p.</i>	6.7 (2.2%)	9.50 (9.51)	83.0 (45.0)	0.768**
<i>Pinus m.</i>	4.1 (1.3%)	7.01 (11.29)	71.5 (80.5)	0.867*
<i>Tsuga h.</i>	2.7 (0.9%)	15.04 (13.05)	85.0 (34.2)	0.725*

to mixed-severity fires and the resulting post-disturbance stand mosaic patterns.

While distance to residual forest constrained post-fire regeneration, intraspecific mechanisms of seed dispersal conferred advantages to ‘fire-evaders’ and early seral species, such as Lodgepole pine, vying for exposed light, space, and soil minerals. This observation prompts further analysis to discern the links between higher regeneration rates and available space versus distance to seed sources in our study site.

4.2. Species responses to beetle outbreaks preceding wildfire events

We explored the efficacy of remote sensing methods combined with

traditional field surveys in detecting Mountain pine beetle outbreaks and wildfire burn severity in our study. Our results affirmed the accuracy of these measures for detecting MPB attacks, with an Overall Accuracy score of 82.4% that was consistent with estimates from studies by [Wulder et al. \(2006b\)](#) and [White et al. \(2005\)](#). Moreover, accuracy of measures for detecting wildfire burn severity classed were also consistent with previous studies ($>87\%$ OA; [Schepers et al., 2014](#)).

Amidst the ongoing discourse on climate warming trends heightening risks of compounded disturbances between wildfires and beetle outbreaks ([Bigler et al., 2005](#); [Dale et al., 2001](#); [Kulakowski et al., 2013](#); [Paine et al., 1998](#)), our findings contribute novel evidence regarding the temporal dynamics of these disturbance partnerships. Notably, red-stage beetle outbreaks had a detrimental effect on Lodgepole pine seedling patterns, whereas grey-stage outbreaks showed no significant negative associations ($\beta = 0.173$). Intriguingly, low severity fire produced significantly positive benefits ($\beta = 0.168$, $p < 0.000$). The observed variability in seedling patterns across grey and red-stage outbreaks and different fire severities implies a weakening impact of beetle-caused tree mortality on the ecological consequences of subsequent wildfire events over time. This aligns with previous studies suggesting that such outcomes relate to the gradual decay and loss of surface fuels induced by wind or climate since the initial attack ([Hicke et al., 2012](#); [Talucci et al., 2019a](#); [Talucci and Krawchuk, 2019](#)).

Given the significance of topography and wind, our models suggest that wildfire and beetle disturbances should be considered in conjunction with site-level environmental factors. Research suggests that in such exposed sites of these subalpine and montane forests, topographic features only serve to intensify climatic shifts in growing conditions and disturbance regimes ([Badger et al., 2014](#); [Littlefield, 2019](#)). This

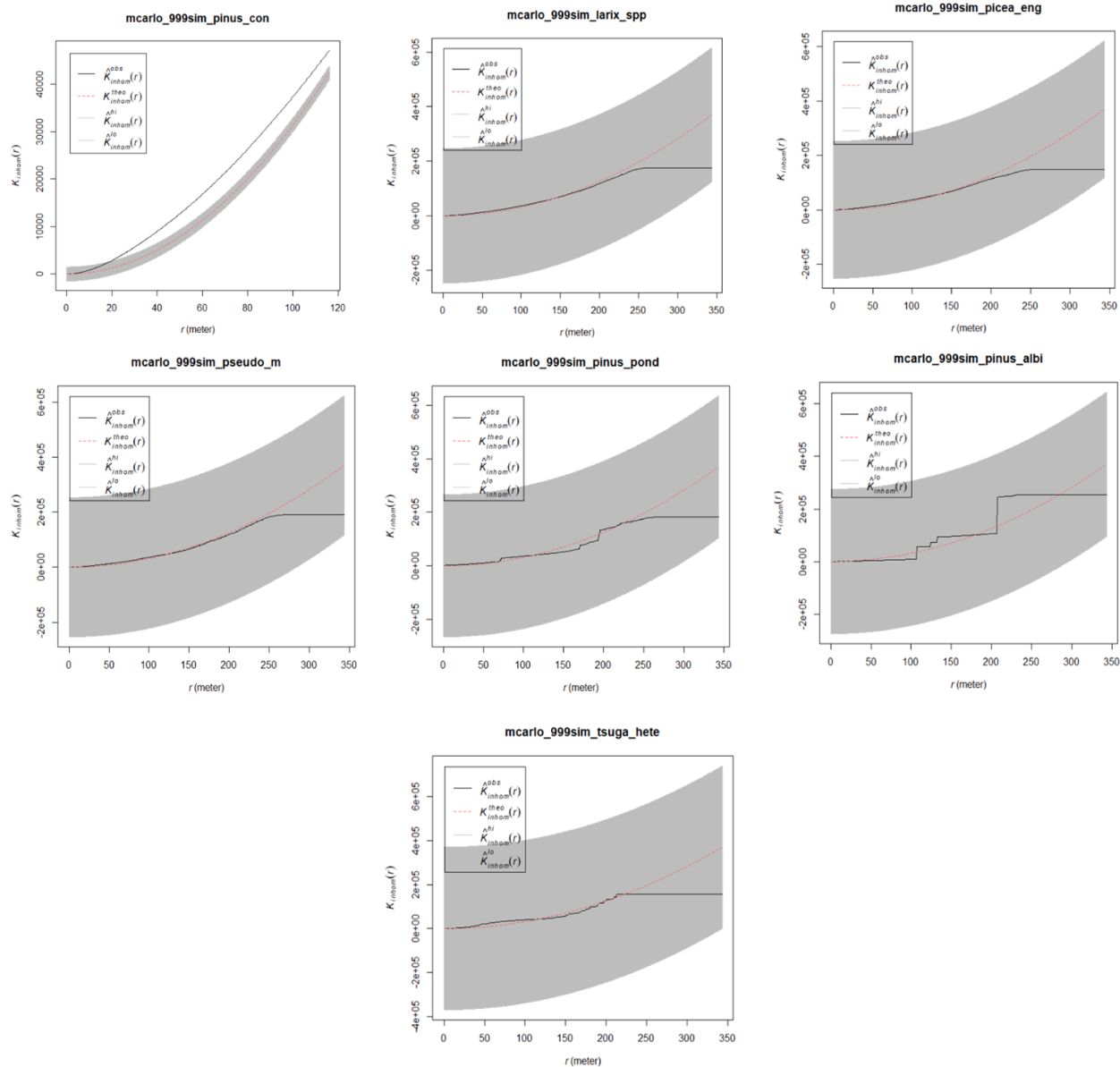


Fig. 6. Ripley's K-diagrams of regeneration patterns of each conifer species across all plots. Clockwise from top left: *Pinus contorta*, *Larix occidentalis*, *Picea engelmannii*, *Pseudotsuga menziesii*, *Pinus ponderosa*, *Pinus monticola*, and *Tsuga heterophylla*.

highlights the particular significance of topography to seed dispersion among serotinous Lodgepole pine stands where prevailing winds may impede or benefit processes of seed rain. Furthermore, such processes are especially important in sites of mixed-severity fires, where aerial dispersal provides a critical seed source to burn patches at lower elevations that are otherwise separated from residual parent trees (Malone et al., 2018). Important to note, however, this study was limited in its analysis of serotinous versus non-serotinous processes. To offer insight on these issues, further data collection is needed to investigate processes of seed production. This calls for future research to investigate such potential links between biotic and abiotic drivers of seed delivery and seedling establishment across varying scales.

5. Conclusions

This study provides fine-scale evidence that compounded disturbances from mountain pine beetle outbreaks and mixed-severity wildfire produce divergent regeneration trajectories among conifer species in the southern Selkirk Mountains of British Columbia. Inhomogeneous Cox process models revealed that red-stage beetle outbreaks significantly reduced lodgepole pine seedling intensity, while grey-stage outbreaks showed no significant effect, suggesting that the ecological consequences of beetle-fire interactions weaken as surface fuels decay over time. Fire severity and distance to seed source governed regeneration patterns for most conifers, with high severity fire disproportionately reducing seedling intensity among fire-avoiders and fire-resisters. Topography and wind emerged as important abiotic mediators of post-fire seedling establishment, particularly for wind-dispersed species

Table 12

Regression estimates of seedling intensity and their influence by covariates according to inhomogeneous Cox-process models.

	Int.	SS-dist	Aspect	RIX	Wind	MPB-G	MPB-R	Fire-H	Fire-L
All seeds	−4.250*** 0.001	−0.012*** 0.001	0.554*** 0.001	0.626*** 0.041	0.272*** 0.001	−0.001 0.990	−0.309*** 0.001	0.207** 0.001	0.395*** 0.001
Pinus	−4.560*** 0.001	−0.012*** 0.001	0.550*** 0.001	0.650*** 0.067	0.277*** 0.001	0.043 0.465	−0.375*** 0.001	0.238** 0.001	0.383*** 0.001
contorta	−5.862*** 0.001	−0.017*** 0.001	0.640*** 0.001	0.658 0.531	0.300* 0.008	−0.160 0.391	−0.106 0.590	0.676** 0.001	0.369** 0.007
Larix	−6.624*** 0.001	−0.011*** 0.001	0.409*** 0.007	1.473 0.187	0.127 0.288	−0.107 0.583	0.115 0.568	−0.069 0.751	0.437** 0.001
occidentalis	−7.030*** 0.001	−0.010*** 0.001	0.567** 0.002	−0.189 0.885	0.320* 0.022	−0.365 0.124	0.114 0.649	−0.277 0.319	0.492** 0.001
Picea	−3.825** 0.020	−0.008** 0.004	0.516 0.122	−0.495 0.843	−0.579* 0.023	0.545 0.143	−0.253 0.535	−0.734 0.115	0.282 0.247
ponderosa.	−26.379*** 0.001	−0.008 0.093	1.984*** 0.001	0.350 0.898	3.506*** 0.001	−1.083 0.098	−0.103 0.867	1.626** 0.005	1.378*** 0.001
Pinus	−13.249*** 0.001	−0.004 0.426	0.686 0.178	7.925* 0.022	0.254 0.489	0.792 0.074	−1.568** 0.009	−1.205 0.135	0.420 0.276
monticola									
Tsuga heterophylla									

such as Western white pine. As climate change shortens beetle-fire disturbance intervals, the observed species-specific responses suggest that pioneering lodgepole pine cohorts may face compounding regeneration deficits, with implications for successional outcomes and forest resilience across western North American conifer forests. Future research should examine serotinous versus non-serotinous seed delivery processes and their interactions with topographic and climatic gradients at broader spatial scales.

CRedit authorship contribution statement

Lauren K. Banks: Writing – review & editing. **John Wilson:** Validation, Investigation. **Adrian Leslie:** Resources, Investigation. **Seamus Murphy:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The author(s) declare that no generative AI or AI-assisted technologies were used in the preparation, writing, or editing of this manuscript.

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Canada, is situated within the traditional territories of the Ktunaxa, Syilx, and Sinixt Peoples in the West Kootenay region of British Columbia.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.foreco.2026.123985](https://doi.org/10.1016/j.foreco.2026.123985).

Data availability

Research Link Provided

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